

# MATH 215 SI

## Session 2

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### Lectures 5 - 8

- ▶ 12.6 Cylinders and Quadric Surfaces
- ▶ 13.1 Vector Functions and Space Curves
- ▶ 13.2 Derivatives and Integrals of Vector Functions
- ▶ 13.3 Arc Length and Curvature
- ▶ 13.4 Motion in Space: Velocity and Acceleration

## 12.6 Cylinders and Quadric Surfaces

A cylinder is a surface that consists of all lines (rulings) parallel to a given line and passing through a given plane curve.

- ▶ If one variable is missing from an equation of a surface, then the surface is a cylinder with rulings parallel to that variable's axis
- ▶ A quadric surface is the graph of a second-degree equation in  $x, y, z$
- ▶ To identify a quadric surface, find its traces (intersections with planes parallel to the coordinate planes)

## 12.6 Cylinders and Quadric Surfaces

cont.

The Standard Quadric Surfaces

- ▶ Ellipsoid:  $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$
- ▶ Elliptic paraboloid:  $\frac{z}{c} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$
- ▶ Hyperbolic paraboloid  $\frac{z}{c} = \frac{x^2}{a^2} - \frac{y^2}{b^2}$
- ▶ Cone:  $\frac{z^2}{c^2} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$
- ▶ Hyperboloid of one sheet:  $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$
- ▶ Hyperboloid of two sheets:  $-\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

# Exercises

## 12.6 Cylinders and Quadric Surfaces

- ▶ What surface in  $\mathbb{R}^3$  is represented by the equation  $z = x^2$ ?
- ▶ Identify the surface given by  $y = x^2 - z^2$  by finding its horizontal and vertical traces (with respect to  $y$ )

## 13.1 Vector Functions and Space Curves

A vector function  $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$  assigns a vector to each value of the parameter  $t$

- ▶ A vector function is a function whose domain is a set of real numbers and whose range is a set of vectors.
- ▶  $\lim_{t \rightarrow a} \mathbf{r}(t) = \left\langle \lim_{t \rightarrow a} f(t), \lim_{t \rightarrow a} g(t), \lim_{t \rightarrow a} h(t) \right\rangle$ , provided the limits of the component functions exist
- ▶  $\mathbf{r}$  is continuous at  $a$  if  $\lim_{t \rightarrow a} \mathbf{r}(t) = \mathbf{r}(a)$
- ▶ A space curve  $C$  is the set of all points  $(f(t), g(t), h(t))$  for  $t \in I$  for some interval  $I$ .

## 13.2 Derivatives and Integrals of Vector Functions

The derivative of a vector function is given componentwise

$$\mathbf{r}'(t) = \lim_{h \rightarrow 0} \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h} = \langle f'(t), g'(t), h'(t) \rangle$$

- ▶  $\mathbf{r}'(t)$  is the tangent vector to the curve at the point defined by  $\mathbf{r}(t)$ , provided  $\mathbf{r}'(t)$  exists and  $\mathbf{r}'(t) \neq \mathbf{0}$
- ▶ Integration is also componentwise:  $\int \mathbf{r}(t) dt = \langle \int f(t) dt, \int g(t) dt, \int h(t) dt \rangle$

## 13.2 Derivatives and Integrals of Vector Functions

cont.

### Differentiation Rules

Let  $\mathbf{u}$ ,  $\mathbf{v}$  be differentiable vector functions,  $c$  a scalar, and  $f$  a scalar function.

$$\blacktriangleright \frac{d}{dt}[\mathbf{u}(t) + \mathbf{v}(t)] = \mathbf{u}'(t) + \mathbf{v}'(t)$$

$$\blacktriangleright \frac{d}{dt}[c\mathbf{u}(t)] = c\mathbf{u}'(t)$$

$$\blacktriangleright \frac{d}{dt}[f(t)\mathbf{u}(t)] = f'(t)\mathbf{u}(t) + f(t)\mathbf{u}'(t)$$

$$\blacktriangleright \frac{d}{dt}[\mathbf{u}(t) \cdot \mathbf{v}(t)] = \mathbf{u}'(t) \cdot \mathbf{v}(t) + \mathbf{u}(t) \cdot \mathbf{v}'(t)$$

$$\blacktriangleright \frac{d}{dt}[\mathbf{u}(t) \times \mathbf{v}(t)] = \mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t)$$

# Exercises

## 13.1 Vector Functions and Space Curves

## 13.2 Derivatives and Integrals of Vector Functions

The trajectory of a particle is given by  $\mathbf{r}(t) = \langle \sqrt{3}t^2, 2t^3, \sqrt{6}t^2 \rangle$ . Let  $C$  denote the corresponding space curve.

- Find an equation for the tangent line to  $C$  at the point  $(4\sqrt{3}, 16, 4\sqrt{6})$

## 13.3 Arc Length and Curvature

- The length of a space curve  $\mathbf{r}(t)$ ,  $t \in [a, b]$  is given by:

$$L = \int_a^b |\mathbf{r}'(t)| dt = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2 + [h'(t)]^2} dt$$

- The curvature of a curve is defined by  $\kappa(t) = \frac{|\mathbf{T}'(t)|}{|\mathbf{r}'(t)|}$  or (more usefully):

$$\kappa(t) = \frac{|\mathbf{r}'(t) \times \mathbf{r}''(t)|}{|\mathbf{r}'(t)|^3}$$

# Exercises

## 13.3 Arc Length and Curvature

- Recall the last exercise. Now find the length of  $C$  on the interval  $t \in [0, \sqrt{8}]$ .

## 13.4 Motion in Space: Velocity and Acceleration

If a particle's position at time  $t$  is  $\mathbf{r}(t)$ , then

- ▶ Velocity:  $\mathbf{v}(t) = \mathbf{r}'(t)$
- ▶ Speed:  $|\mathbf{v}(t)| = |\mathbf{r}'(t)|$
- ▶ Acceleration:  $\mathbf{a}(t) = \mathbf{v}'(t) = \mathbf{r}''(t)$

# Exercises

## 13.4 Motion in Space: Velocity and Acceleration

In this problem all coordinates are measured in meters and time is measured in seconds. At time  $t = 0$  a ladybug, named Sam, is at position  $(1, 1, 1)$  and is flying with constant velocity  $\langle 1, 2, 3 \rangle$  meters per second. A sensor placed at  $(3, 6, 7)$  can detect ladybug motion that occurs within a sphere of radius 7 meters. Does the sensor detect Sam? If so, at what time is Sam last detected by the sensor?

Fin

This concludes session 2.